SEKED / MOSCOW PAPYRUS 14 / SAVED READING

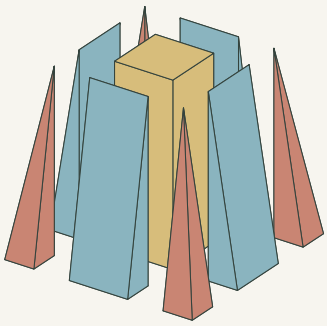
A pyramid without its point.

The surviving example has lower side 4, upper side 2, and height 6. Its calculation gives 56. This reading preserves that example separately from your selected experiment.

Your experiment

Lower side 4, upper side 2, height 6. Spread 80/100; turn 35°; selected group all; cut 50% of the height. All lengths have one common chosen unit.

THE SAME PIECES / TRANSLATED APART

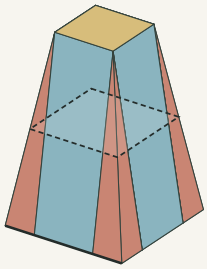


Lower side 4 · upper side 2 · height 6

FIXED DRAWING SCALE / DEPTH FORESHORTENED

2 units

9 NONZERO PIECES / ASSEMBLED



Lower side 4 · upper side 2 · height 6

FIXED DRAWING SCALE / DEPTH FORESHORTENED

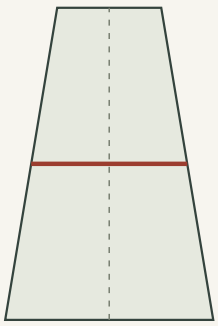
2 units

An assembled and a separated view of the same ideal solid. Moving pieces changes their positions, not their volumes. Vanishing pieces are omitted at a prism or pyramid boundary.

Volumes in cubic units

Piece	One	Count	Together
Central prism	24	1	24
Side wedge	6	4	24
Corner pyramid	2	4	8
Total			56

VERTICAL SECTION / SAME SCALE ON BOTH AXES



Cut height 3 / 6

Square side 3 · area 9

The cut's square side changes linearly with height. Its area is the square of that side. A profile is drawn with equal horizontal and vertical scales.

Why nine pieces add to the rule

Call the lower side *a*, the upper side *b*, and the vertical height *h*. Write $d = a - b$. Partition the solid with the four vertical planes through the upper square's edges.

1. The middle is a square prism: $\mathbf{b^2h}$.
2. Each side piece is a triangular prism. Its right-triangle section has base d/2 and height h, and its length is b: **bdh/4**. Four give **bdh**.
3. Each corner piece is a square pyramid, with base side d/2 and height h: $\mathbf{d^2h/12}$. Four give $\mathbf{d^2h/3}$.

$$V = h[b^2 + b(a - b) + (a - b)^2/3]$$
$$= h(a^2 + ab + b^2)/3.$$

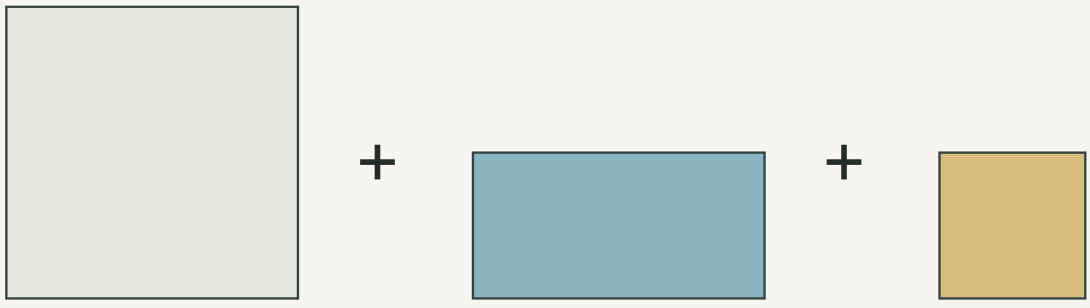
The pyramid volume theorem (one third of base area × perpendicular height) is an input to this explanation. This dissection does not independently prove that theorem. The corner pyramids have an apex over one base corner; they are oblique, with the same perpendicular height h.

A modern dissection of an ideal square frustum: parallel, aligned, centred square ends; straight edges; one common unit. It explains a volume identity. The retained papyrus example does not record this dissection or establish how the rule was discovered. No monument, material, building method or capacity in litres is inferred.

Two tempting estimates

Average the end areas, then multiply by height: **60**, an excess of 4. Square the middle side, then multiply by height: **54**, a shortfall of 2. Exact volume: **56**. All three agree for a prism; the first two miss a tapered solid.

THREE PRODUCTS / ONE COMMON LENGTH SCALE



The product sum 28 is not the area of a horizontal slice. Multiplying its third by the height gives the volume.

The recorded calculation: 4, 2, 6

English quotations below are from Touraeff's 1917 translation, printed p. 101. They are not new translations from the Egyptian. Your selected reading step was 2.

1. ### Square the lower side

   > square this 4, which gives 16

   $$4 \times 4 = 16$$

   The lower square has side 4. Its area is 16 square units. A square side and a square area have different dimensions.

2. ### Double four

   > duplicate 4, which gives 8

   $$2 \times 4 = 8$$

   This is the wording of Touraeff's 1917 translation. In this example the upper side is 2, so doubling 4 also equals the product of the two sides. Applying a × b to other dimensions is a generalization, not another numerical example on this page.

3. ### Square the upper side

   > square the 2, which gives 4

   $$2 \times 2 = 4$$

   The upper square has side 2 and area 4. The three area products are now 16, 8 and 4. The middle product 8 is not the area of the horizontal slice halfway up: that slice has side 3 and area 9.

4. ### Add the three products

   > Add the 16 to the 8 and the 4, which gives 28

   $$16 + 8 + 4 = 28$$

   These three products are added. The result 28 is larger than the lower square's area 16; it is not the area of any one horizontal section of this solid.

5. ### Take a third of the height

   > take ⅓ of 6, which gives 2

   $$6 \div 3 = 2$$

   The vertical height is 6. One third is 2. The printed translation's opening contains an ellipsis; the 6 is explicit in this operation and in the author's following explanation.

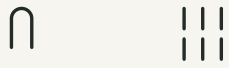
6. ### Reckon with the sum twice

   > take 28 twice, which gives 56

   $$28 \times 2 = 56$$

   The final number is a volume: 56 cubic units in the square-frustum interpretation. This is the surviving numerical calculation. The nine-piece dissection is a separate modern explanation of why a general rule works.

Reading the number signs



16 = 10 + 6



56 = 50 + 6

The drawing above is a modern schematic of additive tens and units, not a tracing. Compare 16 and the final 56 in the typeset transcription on p. 100. Grouping and orientation in the printed page differ. Hieroglyphic transcription, cursive hieratic writing, and a modern Arabic numeral are separate representations.

Complete printed evidence

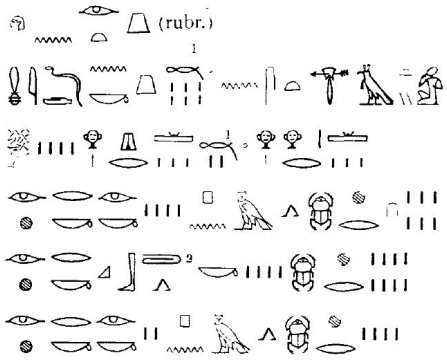
B. Touraeff, *The Volume of the Truncated Pyramid in Egyptian Mathematics*, *Ancient Egypt* (1917), pp. 100–102. This preliminary publication precedes Struve's complete 1930 edition. Its older column labels are retained. No surviving dissection is supplied here.

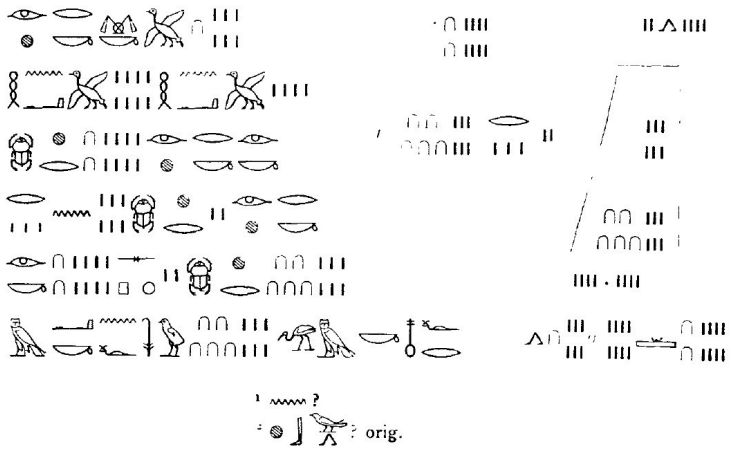
Printed p. 100: The typeset Egyptian transcription. Touraeff's editorial transcription of columns XX–XXI, including the figure and numerical working. These are printed hieroglyphs, not the original cursive marks. The preliminary paper's counts and numbering predate the complete 1930 edition. Digitized original. PDF p. 115; Archive leaf 114; SHA-256 e821f2e2e216f87bbeef0b05c871943214ebc8f9803074378b283202f42b635f.

THE VOLUME OF THE TRUNCATED PYRAMID IN EGYPTIAN MATHEMATICS.

In the collection formerly belonging to Mr. Golenistsheff and recently acquired by the Museum of Fine Arts in Moscow, there is a Mathematical Papyrus of thirty-six columns, in hieratic writing of the epoch of the late Middle Empire. Paleographically it is like some of the Illahun papyri, whilst the breadth of its leaves brings it near to the MSS. of Sinuhe, found in the Ramesseum. This papyrus contains nineteen problems, some of which give us new types of calculation unknown till now, and therefore somewhat difficult to comprehend. Four of these problems are geometrical ones. The first shows how to define the length of the sides of a quadrilateral, when the relation of the sides and the area of the quadrilateral are known. The two next give a method of calculating the area of a triangle: a method already known to us. The fourth presents us, I am inclined to think, with something altogether new in Egyptian scientific literature.

This is the transcription of the XXth and XXIst columns of the papyrus:—

XX. 

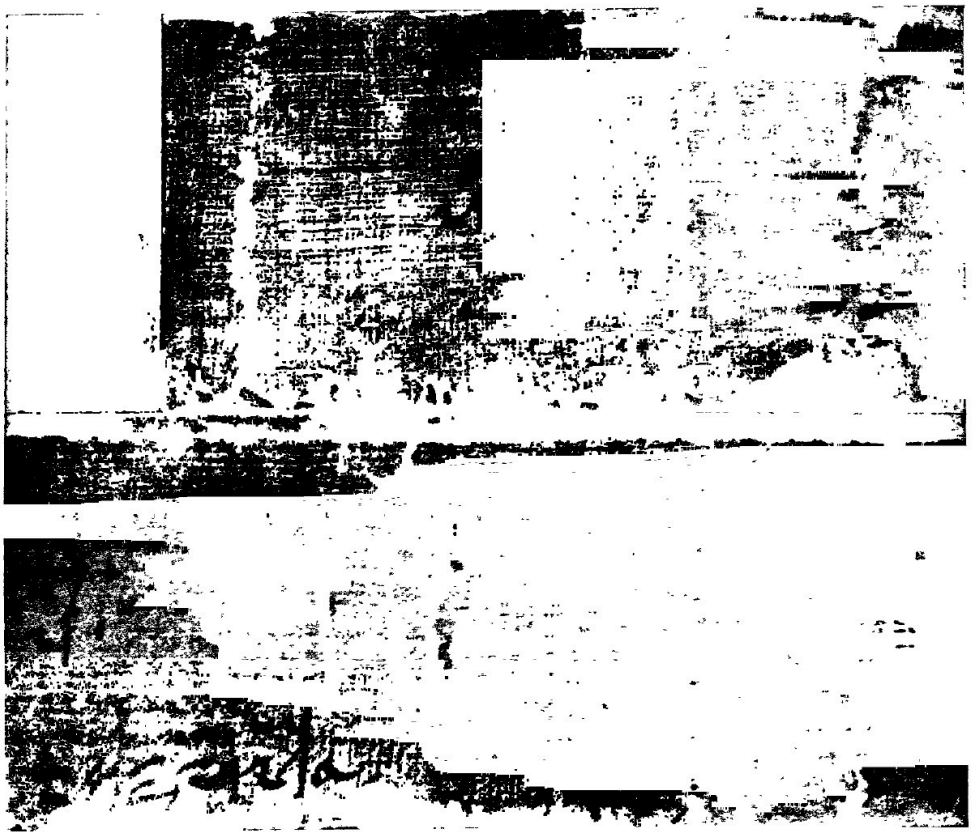
XXI. 

Printed p. 101: Translation, photograph and interpretation. Read the short translation at the top, then the author's modern formula below the photograph. The photograph has severe tonal loss in this digitization. No part of it has been redrawn or retouched. The app makes no new decipherment of its marks. Digitized original. PDF p. 116; Archive leaf 115; SHA-256 47e1aee3ed1e2db32b483bf02ba131d1bc2010dec24be0a07d8efc3a8e5aec5a.

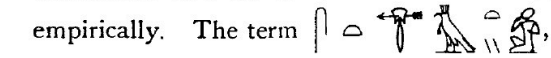
This is the translation of the problem :—

"The problem is to make a ⏢ . If it be said: '. . . . 4 below, 2 above,' do as follows: square this 4, which gives 16; duplicate 4, which gives 8. Do as follows: square the 2, which gives 4. Add the 16 to the 8 and the 4, which gives 28. Do as follows: take $\frac{1}{3}$ of 6, which gives 2. Do as follows: take 28 twice, which gives 56. This is the 56. You will find it correct."

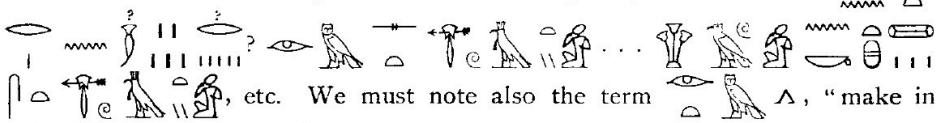
Leaving aside for a while some rather obscure points in our problem, let us try to define its main object. The sketch represents a trapezium, but the calculations do not answer to our definition of the area of a trapezium, nor to the



CALCULATION OF VOLUME OF TRUNCATED PYRAMID.
XIIth dynasty.

Egyptian one. On the contrary, they wholly correspond to our calculations of the volume of a truncated pyramid. Our formula for it: $\frac{h}{3}(A + A_1 + \sqrt{AA_1})$, where A and A_1 represent the area of the upper and under base and h is the height, wholly corresponds to the solution of the Egyptian problem. In fact, the area of the upper base in the latter is $2 \times 2 = 4$, the area of the under base is $4 \times 4 = 16$; $\sqrt{4 \times 16} = 8$; h is 6. Placing these numbers in our formula we get: $\frac{6}{3}(4 + 16 + 8) = 56$. The Egyptian mathematician follows the same method; we only do not understand how he came to define the mean proportional area; it can be only empirically. The term  , which we met here for the first time,

Printed p. 102: The author's conclusion; an editor's note. The bracketed note is the journal editor's intervention. It distinguishes the product of the two sides from the square root of their area product. The author's suggestion that the rule was found empirically on p. 101 is conjectural; no derivation survives in this example. Digitized original. PDF p. 117; Archive leaf 116; SHA-256 6c2345cbb8ab8b131a65adc4d857a3738fee341600cfbe33f9a8e4831733d56a.

is also incomprehensible, and its occurring in some other passages of our papyrus does not add much to its explanation, such as the passage in XXVII:  , etc. We must note also the term , "make in going," for squaring a number.

If only our explanation of the problem be right, we have here a new and interesting fact, *i.e.*, the presence in Egyptian mathematics of a problem that is not to be found in Euclid.

B. TOURAEFF.

[In comparing the Egyptian with the modern formula, where we write $\sqrt{AA_1}$ or $\sqrt{aa\,bb}$, the Egyptian took the simpler form $a \times b$.—ED.]

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B. Touraeff, “The Volume of the Truncated Pyramid in Egyptian Mathematics”,
Ancient Egypt (1917), pp. 100–102.

Digitized volume: Internet Archive in.ernet.dli.2015.107209, supplied by the
Digital Library of India. These are complete printed pages, rendered without
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Original 1917 publication: public domain. Touraeff (Boris Turaev) died in 1920.
The paper is a preliminary publication, earlier than Struve's complete 1930
edition. Its column counts and numbering are retained as historical statements,
not used to identify the modern problem number. Its “empirically” is the
author's conjecture about derivation, not a fact established by the papyrus.

The author's English translation, numerical example, hieroglyphic transcription,
photograph, comments and the bracketed editor's note have distinct roles.
The interactive dissection and all generalized algebra are modern teaching
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these pages. “Problem 14” is the later conventional numbering.

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